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Software engineer with a research interest in applied LLMs and natural language processing. Recipient of the ICAICTA 2024 Best Paper Award for fine-tuning Llama 2 on sarcasm detection. Currently pioneering AI integration at CSG, shipping LLM-powered systems with prompt engineering, confidence-thresholded inference, and human-in-the-loop safety. Seeking graduate study to pursue rigorous research in AI/ML.

Education

Institut Teknologi Bandung

August 2020 - October 2024

Bachelor of Engineering in Informatics

CGPA: 3.76/4.00 (Cumlaude)

Relevant Coursework: Algorithm and Data Structure, Software Project Management, Database

- Management, Platform-based Application Development, Parallel and Distributed Systems, Advanced Machine Learning

Publications & Awards

Sarcasm Detection Using Fine-tuned Llama 2

Singapore

International Conference on Advanced Informatics (ICAICTA 2024)

September 2024

Best Paper Award. Fine-tuned Llama 2 for sarcasm detection on a curated dataset of **21,000+**

- **samples, outperforming prior benchmarks by 8%;** explored prompt-engineering and parameter-efficient fine-tuning trade-offs to improve nuanced language understanding.

Work Experience

CSG International

Jakarta

Software Developer Engineer II

April 2026 - Present

CSG is a global provider of customer experience, billing, and payment solutions for the telecom industry.

Pioneered AI integration at company by building **AI Email Interpreter** in Java with **LangChain on self-hosted Ollama (qwen3-8b)** that converts raw multilingual client emails into structured order-status-update payloads, with **few-shot prompting, confidence-thresholded inference, and human-in-the-loop fallback** for safety; delivered end-to-end PoC in **3 days for a client hackathon** and was **showcased at MWC Barcelona 2026** by pre-sales as the flagship order-to-inbox capability.

Won 2nd place in internal AI competition by architecting an **AI-Assisted API Transformation engine** that converts natural-language instructions into transformation scripts, **substantially reducing manual API mapping effort** for the customer onboarding workflow.

Delivered AI Analytics PoC that exposes the production database to business users via an **MCP-backed conversational interface**, enabling self-service exploration of operational data without writing SQL.

Lead developer for Banking & Financial Solutions (BFS), driving **10+ international BFS demos including Sibos**, owning microservice configuration in the demo environment and acting as **first-line on-call** for incident triage and resolution.

Shipped supporting AI tooling including a **PDF reader that extracts client financial statements into analytics-ready tables** and an **MCP server orchestrating API operations** through a unified framework, accelerating subsequent BFS demo turnaround.

Software Developer Engineer I

January 2025 - April 2026

Contributed to development that earned company TMForum DTW 2025 award by architecting innovative engine for **intent-driven network API transformations** as MVP of Catalyst project in **collaboration with Ericsson**, demonstrating cutting-edge telecom automation capabilities.

Achieved zero transaction failures by implementing comprehensive end-to-end recovery mechanism with **idempotency features, process engine auto-retry, and enhanced internal error handling**, ensuring **100% transaction reliability and recovery** for critical business operations.

Delivered ready-to-go product solutions by improving **50 customized XML product specifications**, refining core business processes, and modernizing Java microservices, **enhancing client configuration efficiency**.

Paper.id

Jakarta

Software Engineer

November 2023 - December 2024

Paper.id is a platform providing invoicing, payment, and financing solutions for businesses.

Accelerated message template creation from hours to under a minute by integrating third-party SMS and WhatsApp broadcast messaging services with **concurrent Goroutines processing**, enabling scalable communication capabilities for business operations.

Architected unified payment engine by developing generalized solution that handles **diverse bank behaviors and merchant-specific requirements**, enabling seamless multi-party integrations while **maintaining system consistency across multiple financial partnerships**.

Achieved 99% transaction success rate by implementing functionality for QRIS payment, direct transfers, virtual accounts, and cross-border remittances compliant with **Indonesia's Central Bank standards**, contributing to Total Payment Volume (TPV) by **4 billion IDR** in a month.

Enhanced system robustness by implementing **dynamic table partitioning and automated transaction reconciliation** with RabbitMQ scheduling and asynchronous transaction callbacks, **increasing query performance by 20%** and **enabling instant execution of reconciliation report generation**.

Suitmedia

Bandung

Software Engineer Intern

May 2023 - September 2023

Suitmedia is a digital agency specializing in technology and marketing, providing solutions and strategic partnerships.

Delivered comprehensive CMS solution by designing and implementing **35 dynamic menus and 50 API endpoints** using Laravel framework with efficient database schemas inferred from Figma prototypes, **ensuring seamless front-end integration** and enhanced website functionality for client requirements.

Optimized system performance and content delivery by redesigning database structures and implementing dynamic content management with static page integration, **resulting in**

improved query performance by 30% and enhanced client usability across both dynamic and static website pages.

Synpulse

Jakarta

Software Engineer Intern

October 2022 - December 2022

Synpulse is a management consulting firm serving financial services providers.

Streamlined core banking system migration by authoring functional requirements for **20**

- **system specifications**, mapping existing to new API specifications and building system flowcharts that kept **Banking Client's Avaloq transition project on schedule**.

Eliminated manual daily data processing by developing automated shell scripts for Excel-to-

- Oracle database conversion, **reducing reconciliation report processing from 15 minutes to instant execution** and ensuring seamless core banking system migration for **Banking Client**.

Technical Skills

- **Programming Languages:** Java, Golang, PHP, Python, JavaScript, Shell Scripting
- **Frameworks & Libraries:** Laravel, React, ViteJS, NodeJS, Fastify, JAX-WS
- **Databases:** PostgreSQL, MySQL, ArangoDB, Oracle Database
- **Tools & Platforms:** RabbitMQ, Kafka, OpenTelemetry, Vault, Consul, Retrofit, Figma
- **Cloud & DevOps:** Google Cloud Platform, AWS, Docker, Jenkins
- **AI / ML:** LangChain, Ollama, OpenRouter, Llama 2, Qwen, Model Context Protocol (MCP), Prompt Engineering, LLM Evaluation